

Program Pembangunan
Olimpiad Matematik (Siri 1)

Combinatorics

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1 Warm-Up

1.1 Theory

- Basic Counting
- Complementary Counting
- Principle of Inclusion and Exclusion

1.2 Problems

1. If a set S has n elements, find the number of subsets of S .
2. How many ways can you seat 6 people around a circular table, if rotations count as one?
3. How many ways can you seat 7 girls and 3 boys in a row such that no boy sits next to another boy?
4. Find the number of ways to rearrange the letters of the word

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5. How many four-digit positive integers contain at least one even digit?
6. How many positive integers less than 100 are divisible by 3 or 5?
7. How many positive integers less than 300 are divisible by 3, 4 or 5?

2 Pigeonhole Principle

2.1 Theory

If $n + 1$ or more pigeons are placed into n holes, one hole must contain more than one pigeon.

Generalised: If m pigeons are placed into n holes, one hole must contain at least $\lceil \frac{m}{n} \rceil$ pigeons.

2.2 Problems

1. Prove that among 400 people, there must be two of them who share the same birthday.
2. Suppose you have an infinite number of blue socks, red socks and yellow socks in a drawer. What is the minimum number of socks you must pull out from the drawer (assuming you are not looking) to guarantee there is a pair of the same color?
3. Prove that in a group of n people, there are two who have the same number of friends (assuming friendship is mutual).
4. Suppose S is a set of $n + 1$ integers. Prove that there exists distinct a, b in S such that $a - b$ is a multiple of n .
5. Five points are placed inside an equilateral triangle with sides of length 2. Prove that the distance between some pair of points is at most 1.
6. Let $S = \{1, 2, 3, \dots, 2n\}$. Show that if we choose $n + 1$ numbers from S , then there exist two numbers such that one is a multiple of the other.
7. Show that for any positive integer n , there exists a positive multiple of n that only contains the digits 1 and 0.
8. Prove that there exists a subset of n (not necessarily distinct) integers whose sum is divisible by n .

3 Problems

1. (JBMO SL 2013 C1) Find the maximum number of different integers that can be selected from the set $\{1, 2, \dots, 2013\}$ so that no two exist that their difference equals to 17.
2. (Manhattan MO 2005) Prove that having 100 whole numbers, one can choose 15 of them so that the difference of any two is divisible by 7.
3. (JBMO SL 2012 C1) Along a round table are arranged 11 cards with the names (all distinct) of the 11 members of the 16th JBMO Problem Selection Committee . The cards are arranged in a regular polygon manner . Assume that in the first meeting of the Committee none of its 11 members sits in front of the card with his name . Is it possible to rotate the table by some angle so that at the end at least two members sit in front of the card with their names?
4. (JBMO SL 2008 C3) Integers $1, 2, \dots, 2n$ are arbitrarily assigned to boxes labeled with numbers $1, 2, \dots, 2n$. Now, we add the number assigned to the box to the number on the box label. Show that two such sums give the same remainder modulo $2n$.
5. (JBMO 2007 P3) Given are 50 points in the plane, no three of them belonging to a same line. Each of these points is colored using one of four given colors. Prove that there is a color with at least 130 scalene triangles with vertices of that color.